CS-8TH

Mobile Application Development

CSCP-4063

Program	**BSCS**
Credit Hours	**3**
Duration	**16 Weeks / 32 sessions**
Prerequisites	**Object-Oriented Programming**

P. Tahir

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COURSE REQUIREMENT

This course is for learning the Kotlin, XML, for mobile app development.

COURSE OBJECTIVES

The course will cover beginner level Android Application development using Android Studio. The objective of this course is to introduce students to the necessary concepts of Android Development that can then be used by the students to create Android Applications of their own. The course is designed for students who have a basic understanding of programming concepts and wish to learn professional development of Android Apps. The course is technical in nature, with a strong emphasis on doing hands-on coding assignments in the classroom. Android Studio will be used as the choice of IDE for developing the apps. Kotlin will be the primary language being taught. Throughout the duration of the course, students will work in groups, tackling programming challenges and creating their own apps using state of the art Android Development tools.

LEARNING OUTCOMES

By the end of this course, students will be able to:

- Understand the basic concepts of Android Development
- Use Google's Official tool for Android Development
- Code at ease in Kotlin
- Create any app in Android Studio, given functional requirements
- Create a complete App of their own that will be deployable on Google's play store

TEACHING LEARNING METHOD

- Lectures, Reading Material.
- Guest Speakers from industry.
- Skills Development Exercises.
- Project Report / Term Paper.
- Movies, Handouts, Case Studies and any other Teaching-Learning Tool.

REFERENCE TEXT & BOOKS:

Android Development with Kotlin by Igor Wojda 2017 edition
Head First Android Development: A Brain-Friendly Guide by David Griffiths and Dawn Griffiths 2015 edition

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Calendar of Activities

Weeks	Contents	Activities
1	Introduction to course Division of marks Brief introduction of: • Android OS • Android Studio Linux Vs Android Introduction to Kotlin Language Syntax of Kotlin including: • A basic Kotlin program e.g. main function • Variables • Data types • Conditional statements • Loops: While loop For loop Different variations of for loop in Kotlin • Lists and Arrays in Kotlin • When statement • Class in Kotlin • Data members and member functions • Constructors: Primary constructor Secondary constructor	
2	Introduction to Android studio IDE • Creating First android studio project • Introduction to: • Manifest File • Layouts • Back-end coding e.g. kotlin (.kt) files • Gradle files • Dependencies • Running first android studio project Setting up a Virtual device • Introduction to an Emulator: • Android studio's own virtual device/emulator Using 3rd party emulator e.g. Memu, NoX, or BlueStack • Using physical phone to Run and Test an android app Introduction to Views: • Button • TextView • EditText • Auto-complete TextView	

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	• ListView	
3	• Using 3rd party emulator e.g. Memu, NoX, or • BlueStack • Using physical phone to Run and Test an android • app • Introduction to Views: • Button • TextView • EditText • Auto-complete TextView • ListView • .Creating First Android app • Understanding of click events • Button click listener • Button Long click listener • Displaying a Toast Message • Displaying an AlertDialog Message • A simple Calculator application	Assignment-1
4	• Introduction to Activities • Creating Multiple activities • Passing data to an activity • Shifting among multiple activities • Receiving data in new activity (Passed by previous • activity) • GSM Services • Sending SMS (GSM) using SmsManager API • Sending E-mails • Making a phone call application • Permissions in android • What is a permission • Manifest permission Vs Run time permission • (android SDK >= 6.0) • Writing manifest permission • Writing a run time permission	Quiz-1
5	• Animations • What is an animation • Creating an animation in android • Loading an animation • Notification • What is a notification	

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	• Creating a Notification • Pushing a notification • Associating an activity with a notificatoin • Wi-Fi • How to Enable / Disable Wi-Fi • Scanning and getting a list of availabel Wi-Fi • networks • Bluetooth • How to Enable / Disable Bluetooth • Scanning and getting a list of availabel bluetooth • devices • Camera • How to access Camera • Taking pictures with camera •	
6	• Data Storage • Introduction to different storages in android e.g. • SharedPreferences, SQLite database, MySQ • database, and Firebase database • Creating a sharedPreferences storage • Storing data in sharedPreferences • Retreiving data from SharedPreferences • 12 Activity Life Cycle • Life of an activity • Different call back methods: • onCreate() • onStart() • onResume() • onPause () • onStop() • onRestart() • onDestroy() • Splash Screen • What is a splash screen • Creating a Splash screen •	Assignment-2
7	• ListView • Introduction to ListView in android • Setting up data on a list view • Using custom layout for listview • Customizing adapter for a ListView • Auto-complete TextView • Difference in TextView & Auto-complete	Quiz-2

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	• TextView • Setting up data for Auto-complete text view. • Locations in Android • GPS location Vs Network provided location • Introduction to Latitude and Longitude • Getting GPS location • Progress dialogue • Converting GPS location to Actual/physical address • using GeoCoder API •	
8	• **Revision**	
9	**MID TERM**	
10	• SQLite Database • What is SQLiteDatabase • SQLite Database • What is SQLiteDatabase • SQLite Database • Retreiving data from SQLite Database • Applying different filters on data	
11	• Firebase Database • Introduction to Firebase SDK • Setting up a firebase project • Connecting firebase application with firebase server • Different settings on Firebase Server • Firebase Database • Creating a user with email and password • Signing in to Firebase database using Account • credentials • Storing single data item in Firebase database •	Assignment-3
12	• Firebase Database • Storing complete objects in Firebase database • Introduction to Snapshot • Retrieving data from Firebase • MySQL Database • Introduction to MySQL database • Introduction to hosting	Quiz-3

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	• Introduction to XAMP / WAMP server • Setting up XAMP server • Creating a MySQL database using XAMP server	
13	• MySQL Database • Introduction to PHP language • Syntax of PHP language • Writing PHP script for storing data in MySQL database • Writing PHP script for reading data from MySQL database • JSON encoding • MySQL Database • Connecting an android application with MySQL database using AsyncTask API • Stroing data in MySQL database • Reading data into android App from MySQL database	
14	• Audio Management in Android • Playing an audio file in android • Recording audio in android • Accessing files from ROM and SD card • Threads • Introduction to Thread • Introduction to Runnable • Writing a Thread • Running a code in multiple threads	Assignment-4
15	• Services • What is a service • Comparison of a service and an activity • Creating a service • Running a service • Destroying / stopping a service • Broadcast Receiver • What is a broadcast receiver? • Different broadcast signals in android OS • Registering an application for broadcast signal(s) • Un-registering a broadcast receiver	Quiz-4
16	• Project Evaluation • Revision	
17	**FINAL TERM**	

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